

CSE 360

Individual Homework 3

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Individual Homework 3

1. A Listing of the Five Automated Tests

Test Name	Location	Purpose
Password Test: Valid Password	PasswordEvaluator.java	Confirm password "Aa1!abcd" passes FSM validation
Username Test: Valid Username	UserNameRecognizer.java	Verify username "Test_User . 1" is accepted by the FSM recognizer
GUI Test – FirstPage Button Label	FirstPage.java	Validate that the "Continue" button label displays the correct text
Invitation Code Test	InvitationPage.java	Confirm that generateInvitationCode() returns a 4-character code
Ask Question Submission Test	AskQuestionPage.java	Verify that a question added increases the question list size by one

2. A Description of the Five Automated Test

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```
===== HW3 Automated Tests =====  
Test 1: PasswordEvaluator  
Aa1!abcd  
?  
The password size: 8 | The currentCharNdx: 0 | The currentChar: "A"  
Upper case letter found  
  
Aa1!abcd  
A?  
The password size: 8 | The currentCharNdx: 1 | The currentChar: "a"  
Lower case letter found  
  
Aa1!abcd  
Aa?  
The password size: 8 | The currentCharNdx: 2 | The currentChar: "1"  
Digit found  
  
Aa1!abcd  
Aa1?  
The password size: 8 | The currentCharNdx: 3 | The currentChar: "!"  
Special character found  
  
Aa1!abcd  
Aa1!?  
The password size: 8 | The currentCharNdx: 4 | The currentChar: "a"  
Lower case letter found  
  
Aa1!abcd  
Aa1!a?  
The password size: 8 | The currentCharNdx: 5 | The currentChar: "b"  
Lower case letter found  
  
Aa1!abcd  
Aa1!ab?  
The password size: 8 | The currentCharNdx: 6 | The currentChar: "c"  
Lower case letter found
```

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```
Aa1!abcd
Aa1!abc?
The password size: 8 | The currentCharNdx: 7 | The currentChar: "d"
Lower case letter found
```

```
Input: Aa!abcd
Expected: Valid (empty string), Actual:
PASS
```

Test 2: UserNameRecognizer

Current State	Final State	Input Char	Next State	Date Size
0		T	1	1
1	F	e	1	2
1	F	s	1	3
1	F	t	1	4
1	F	U	2	5
2	F	s	1	6
1	F	e	1	7
1	F	r	1	8
1	F	.	2	9
2	F	1	2	10
1	F	None	1	11

```
The loop has ended.  
Input: Test_User.1  
Expected: Valid (empty string), Actual:  
PASS
```

```
Test 3: FirstPage Title Label
Test 4: InvitationPage Code Generation
Connecting to database...
Connecting to database...
Generated Code: d251
PASS
```

```
Test 5: AskQuestionPage Question Submission
Question added by Tester: Test Question
Before: 0, After: 1
PASS
```

```
===== End of Tests =====
Test: Verify Continue Button Text - Expected: 'Continue', Actual: Continue
Test: FirstPage Loaded - Scene Title: null
Check console output for: 'Test: Verify Continue Button Text'
Test: Continue Button Clicked - Navigating to Admin Setup Page
Test: Admin Setup Page Loaded - Scene Title: Administrator Setup
```

Test 1: testPasswordEvaluator()

Description:

Verifies that the PasswordEvaluator class correctly evaluates a valid password.

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Use the input string "Aa1!abcd" containing uppercase, lowercase, numeric, and special characters with the required length. The test determines if the error message returned is an empty string to determine if the password is valid.

Test 2: testUserNameRecognizer()

Description:

Tests whether UserNameRecognizer can accept a format compliant username.

Enter the string "Test_User.1" containing letters, underscores, dots and numbers that satisfy the state transfer rules defined by the FSM. The test expects the return value to be the empty string, i.e. the username is valid.

Test 3: testFirstPage()

Description:

Tests if the FirstPage interface displays correctly.

Launch the FirstPage scene via JavaFX and check if you can see the header test output printed on the console, e.g. "Test: Verify Continue Button Text". This test verifies that the GUI initialization process is working correctly.

Test 4: testInvitationPage()

Description:

Verifies that the function to generate invitation codes is available in the DatabaseHelper class.

The test calls the generateInvitationCode() method and checks if the generated invitation code is the expected 4 characters. The test passes if the return code length is 4.

Test 5: testAskQuestionPage()

Description:

Tests if the function to add a question in AskQuestionPage works successfully.

Use QuestionsManager.getInstance() to add a question, first record the length of the question list, then add it to verify that the length has increased. Verify that the question is indeed saved.

3. The Link to the HW3 Code in Your Personal Repository

<https://github.com/Benny-Zheng/CSE360/tree/HW3/HW3/FoundationCode/FoundationCode/src/HW3>

4. The Link to the Source for Your Javadoc Inspiration

Reference link: <https://www.oracle.com/java/technologies/javase/javadoc-tool.html>

Explanation:

This example shows how to write professional Javadoc with `@param`, `@return`, `@throws` tags. It clearly describes method usage, parameters, and return values and is suitable for teamwork and open source project maintenance.

https://docs.google.com/document/d/1t8Vm_n0tIZgDMA559IZhbqcf-iO3nnCUem9O5mK43rA/edit?tab=t.0

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5. The Link to Your Screencast

https://asu.zoom.us/rec/share/f7w4MJuJZZ_H7mVdXffX4YD1_g55BFAugj0dAMKonlXg_2G6dOuwugxXeCTbu_k.GqXQOvNDOaEoQnPg?startTime=1743046941000

Password: TqN\$1B?5